Candidate Seat No: ____________________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Bachelor of Science in Information Technology (B.Sc. IT)
SEMESTER-II University Examination May - 2018
CA 408 - Database Fundamentals

Date: 09/05/2018, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 70

Instructions:

1. **Attempt the two sections in separate answer books.**
2. **Figure to the right indicate marks.**
3. **Make suitable assumptions wherever necessary and state them.**

SECTION-I

Q.1 Do as directed : **[11]**

1. In an E-R diagram, Oval is represented by_______________.
   A. entity B. attributes C. link D. relationships
2. σ operator is used to denote the selection operation in relational algebra. **(True/False)**
3. In a relational model, row is termed as ______________.
   A. Tuples B. Relation C. Rows D. Attributes
4. ON DELETE CASCADE option will delete parent as well as child records. **(True/False)**
5. Which of the following can be a multivalued attribute?
   A. Age B. Hobbies C. PAN number D. None
6. External Level must be only one.**(True/False)**
7. Relational Algebra is a ___________ query language that takes two relations as input and produces another relation as output of the query.
   A. Relational B. Structural C. Procedural D. Fundamental
8. In an E-R diagram, Entity is represented by_______________.
   A. rectangle B. circles C. diamond D. ellipse
9. The ____________ operation allows us to find tuples that are common in both relations.
   A. Union B. Intersect C. Minus D. None of above
10. Age attribute is an example of a Derived attribute.**(True/False)**
11. A relationship where two entities are participating is called a _________ relationship.
    A. binary B. ternary C. n-array D. Unary

Q.2 [A] Draw and explain the ANSI/SPARC architecture of the database system. **[5]**
[B] Explain ACID properties of Transaction. **[4]**
[C] Explain Relationship participation constraint using an example. **[3]**

OR

Q.2 [A] Write down the advantages of DBMS. **[5]**
[B] Explain basic operations of relational algebra with example. **[4]**
[C] Explain cardinality of relationship. **[3]**

Q.3 Answer the Following: (Attempt any two) **[12]**

[A] Draw an E-R diagram to indicate attributes, keys and cardinality ratios for below scenario:

- Company organized into the Department. Each department has a unique number with name and location.
- Many employee works in department.
- A company controls a number of Projects. Projects have a unique number, name and a single location.
- Employee name, id, salary, age, and birthdate are recorded.
- An employee is assigned to one department, but may work for one or more Projects.

[B] What are the possible violations for update operation in relation?

[C] Write down the SQL query and convert it into Relational Algebra Query.

Table Structure: Student (Sid, Name, Age, Gender, Programme)

Result (ResultId, Sid, SemId, CGPA, SGPA)

1) Display name of all students who are in B.Sc.(IT) Programme.
2) Display name and CGPA of students who have secured more than 8 CGPA.

[D] List and explain types of attributes in ER diagram. Give an example for each.

SECTION-II

Q.4 Do as directed: **[11]**

1. View is an SQL virtual table that is constructed from other tables. **(True/False)**

2. ____________ Key is used to represent relationships between tables.

A. Primary Key B. Unique Key C. Foreign Key D. Candidate Key

3. Which of the following is a data manipulation command?

A. SELECT B. CREATE C. ALTER D. GRANT

4. SQL stands for Simple Query Language. **(True/False)**

5. A _________ is an SQL statement that appears inside another SQL statement.

A. Index B. Subquery C. Union D. Group By

6. Which SQL keyword is used to retrieve only unique values?

A. distinctive B. unique C. distinct D. different

7. DROP command is used to delete a row from the table. **(True/False)**

8. A function that has no partial functional dependencies is in _______ form.

A. 3NF B. 4NF C. 2NF D. BCNF

9. Which of the following operators cannot be used in a subquery?

A. AND B. < C. IN D. >=

10. Default date format of Oracle is ______________.

A. dd-mm-yyy B. dd-mon-yy C. dd-mm-yyyy D. dd-mm-yy

11. DDL stands for ____________________________.

Q.5 [A] Explain sum(),count(), max(), avg() and min() functions of SQL with suitable example. **[5]**

[B] List out the types of JOIN and explain INNER JOIN with syntax and example. **[4]**

[C] Write a syntax of UPDATE and DELETE commands and explain with suitable example. **[3]**

OR

Q.5 [A] List out conversion functions and explain any two of them using an example. **[5]**

[B] Explain any four Codd's Rules. **[4]**

[C] Write down the syntax of DROP Column and MODIFY Column using ALTER Command. **[3]**

Q.6 Answer the Following: (Attempt any 2) **[12]**

[A] Define Normalization. Explain Normalization upto 3NF using an example.

[B] Write down the syntax to CREATE **Activity** table which has Activityid, Activity_Name, Location, Fee and Campid fields.

Apply the below constraints:

1) Activityid should be primary key.
2) Location must be 'Baroda', 'Surat' or 'Ahmedabad'.
3) Fee cannot be zero.
4) Campid is Foreign key of CampDetail table.

[C] Explain different types of anomalies.

[D] What is VIEW ?Discuss the types of VIEW with suitable example.